

KUSHAGRA PRATAP SINGH

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Summary

Fullstack Software Engineer (B.Tech, Communication & Computer Engineering, LNMIIT, graduating 2027) building production systems across React.js, Node.js/Express, and Python-based ML work. Shipped an LLM-powered feature – Gemini extracting structured fields from freeform text – backed by a pgvector HNSW semantic search index, plus an ecommerce microservice platform (GraphQL, Kafka, Docker, CI/CD). SDE internship experience owning backend features end-to-end – payments, RBAC, uptime monitoring – for a platform serving 7,000+ users. Hands-on with LLM APIs (Gemini, Bedrock/Claude 3), prompt engineering, and vector search; solved 450+ problems across LeetCode, GeeksforGeeks, and Codeforces.

Education

The LNM Institute of Information Technology

B.Tech in Communication & Computer Engineering

Jaipur, Rajasthan

Aug 2023 – May 2027

Aadharshila ‘The School’

Class X – 95.6% Class XII – 96.4%

Apr 2020 – Mar 2022

Experience

Marine Edge Technologies Pvt. Ltd.

Jun 2025 – Nov 2025

Software Development Engineer Intern

Remote

- Implemented a configurable **3-tier RBAC** system and integrated **Razorpay** payments, maintaining **99.9% uptime** across **100+ daily transactions**.
- Owned backend features end-to-end on a scalable, distributed web platform serving **7,000+ users** globally within a cooperative, cross-functional engineering team.
- Improved front-end performance and reliability with **React.js**, cutting load times by **30%** and API error rates by **25%**.
- Deployed **Jitsi Meet** video classes for **500+ concurrent attendees**; added real-time Admin analytics APIs to monitor platform availability and performance.

Projects

Later, Probably | Node.js, PostgreSQL, Prisma, pgvector, Gemini API, React, Recharts | [GitHub](#), [Live](#)

- Shipped an AI task estimation web app where **Gemini** extracts structured fields from freeform text, backed by a **pgvector HNSW** index over **1536-dim embeddings** for fast semantic similarity search.
- Engineered a privacy-first pipeline that pre-aggregates stats server-side before any LLM call, sending **zero raw task data**; delivered **React** + Recharts dashboards tracking predicted-vs-actual duration.

Ecom Microservice GraphQL | Node.js, Express.js, GraphQL, Apollo, MongoDB, Kafka, Docker | [GitHub](#), [Live](#)

- Architected a **3-service** full-stack backend (user, product, order) behind a unified **GraphQL** gateway with **JWT** auth and role-based access control.
- Adopted **Apache Kafka** for event-driven communication, cutting inter-service coupling and enabling independent, reliable scaling; containerized all three services with **Docker** and shipped **CI/CD** via GitHub Actions.

Interlace | Node.js, Express.js, Socket.io, MongoDB, CRDT (RGA), JWT | [GitHub](#), [Live](#)

- Designed a collaborative code editor using a custom **RGA CRDT** built from scratch (no Yjs/Automerger dependency), guaranteeing **Strong Eventual Consistency** across concurrent edits with zero central coordination, backed by an append-only MongoDB ops log with offline queuing and **JWT**-secured rooms.

Skills

- Fullstack Web Development:** JavaScript, Python, HTML, CSS, React.js, Node.js, Express.js, REST API Design, GraphQL.
- AI/LLM Engineering:** Gemini API, Amazon Bedrock (Claude 3), Prompt Engineering, RAG, Vector Search (pgvector, HNSW).
- Backend & Data:** PostgreSQL, MongoDB, Redis, Microservices, Event-Driven Architecture (Kafka), JWT, RBAC.
- Tooling & DevOps:** Git, GitHub Actions, CI/CD, Docker, AWS Lambda, DynamoDB.

Achievements & Certifications

- Solved **450+ DSA problems** across LeetCode, GeeksforGeeks, and Codeforces.
- Launched **CodeFlow AI**, a serverless hackathon app delivering AI-guided learning roadmaps.
- GenAI Workshop** – LNMIIT: full-stack generative AI curriculum, scored **100/100**.
- Lead a **7-10 member** Creative Team for **ACM LNMIIT** Student Chapter, driving design and branding across **2 chapter-wide events**.